

Week 9 Solutions

1. a)

$$R = V/I = \frac{12V}{0.60A} = 20\Omega \quad (1)$$

b) An amount of charge loses a potential energy of $\Delta Q \times V$ as it passes through the resistor.

$$\Delta Q = I \times \Delta t \quad (2)$$

$$\Delta U = (\Delta Q)V = (I\Delta t)V = (0.60A)(60s)(12V) = 430J \quad (3)$$

2. (a) Breaking the wire into segments of length dx , we find the total resistance is

$$R = \int_0^L \frac{\rho}{A} dx \quad (4)$$

$$= \frac{1}{A} \int_0^L \rho_0 e^{-x/L} dx \quad (5)$$

$$= \frac{\rho_0 L}{A} (1 - e^{-1}) \quad (6)$$

(b) According to Ohm's law,

$$j = \frac{E}{\rho} \quad (7)$$

Since the current and area are constant in every cross section of our wire, and the resistivity is a decreasing function of x , E must also be a decreasing function of x .

$$j = \frac{I}{A} \quad (8)$$

$$= \frac{V_0}{R} \frac{1}{A} \quad (\text{because } V_0 = IR) \quad (9)$$

$$= \frac{V_0}{A} \frac{A}{\rho_0 L} \frac{1}{1 - e^{-1}} \quad (\text{using } R \text{ from part a}) \quad (10)$$

$$= \frac{V_0}{\rho_0 L} \frac{1}{1 - e^{-1}} \quad (11)$$

$$E = j\rho \quad (12)$$

$$= \left(\frac{V_0}{\rho_0 L} \frac{1}{1 - e^{-1}} \right) (\rho_0 e^{-x/L}) \quad (13)$$

$$= \frac{V_0}{L} \frac{1}{1 - e^{-1}} e^{-x/L} \quad (14)$$

$$\vec{E} = \frac{V_0}{L} \frac{1}{1 - e^{-1}} e^{-x/L} \hat{x} \quad (15)$$

The electric field (and $\hat{x}$) points from the end at $x = 0$ toward the end at $x = L$.

3. The capacitor can be thought of as a collection of capacitors that have infinitesimal areas and are in parallel. The distance x between the plates of these infinitesimal capacitors is a function of y , the distance from the bottom of the capacitor. In particular, we have

$$x = d + y \tan \theta. \quad (16)$$

Then, the capacitance of one of the tiny capacitors located at a height y is

$$dC = \epsilon_0 \frac{dA}{x} \quad (17)$$

$$= \epsilon_0 \frac{\sqrt{A} dy}{d + y \tan \theta} \quad (18)$$

where we have used $dA = \sqrt{A} dy$. Since all of the capacitors are in parallel, we can simply sum up their capacitances via an integral.

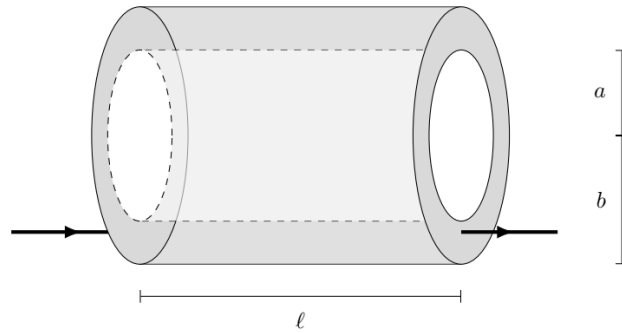
$$C = \int_0^{\sqrt{A}} \epsilon_0 \frac{\sqrt{A} dy}{d + y \tan \theta} \quad (19)$$

$$= \frac{\epsilon_0 \sqrt{A}}{\tan \theta} \ln(d + y \tan \theta) \Big|_0^{\sqrt{A}} \quad (20)$$

$$= \frac{\epsilon_0 \sqrt{A}}{\tan \theta} \ln \left(1 + \frac{\tan(\theta) \sqrt{A}}{d} \right) \quad (21)$$

4. Solution

1. Sketch.



2. Resistance is $R = \rho \frac{\ell}{A}$. In this case, the cross-sectional area is $A = \pi b^2 - \pi a^2$, so

$$R = \frac{\rho \pi \ell}{b^2 - a^2}$$

3. If $\ell \rightarrow 2\ell$, then $R \rightarrow 2R$.

4. If $b \rightarrow c > b$, then

$$R \rightarrow \frac{\rho \pi \ell}{c^2 - a^2} = R_{ca}$$

Note that

$$\frac{1}{R_{bc}} + \frac{1}{R_{ba}} = \frac{c^2 - b^2}{\pi \rho \ell} + \frac{b^2 - a^2}{\pi \rho \ell} = \frac{c^2 - a^2}{\pi \rho \ell} = \frac{1}{R_{ca}},$$

so R_{ca} is made of R_{ba} and R_{cb} in parallel.

5. By Ohm's law,

$$I = \frac{V}{R} = \frac{V(b^2 - a^2)}{\rho \pi \ell}.$$

6. Power dissipated is given by

$$P = IV = \frac{V^2(b^2 - a^2)}{\rho \pi \ell}.$$

5. Submerged resistor with capacitor:

$$V - IR - I_R R = 0 \quad (22)$$

$$V - IR - Q/C = 0 \quad (23)$$

$$I = I_R + I_C \quad (24)$$

Solving for $I_C(t)$:

$$\frac{dI_C}{dt} = -\frac{2}{RC}I_C \implies I_C(t) = I_0 e^{-2t/RC}, \quad I_0 = \frac{V}{R}$$

Current through resistor:

$$I_R(t) = \frac{V}{2R} \left(1 - e^{-2t/RC}\right)$$

Power:

$$P(t) = I_R^2 R$$

Total energy:

$$E(t) = \frac{V^2}{4R} \int_0^t \left(1 - e^{-2t'/RC}\right)^2 dt' = \frac{V^2}{4R} \left(t + \frac{RC}{4} \left(4e^{-2t/RC} - e^{-4t/RC} - 3\right)\right)$$

Temperature increase assuming all energy absorbed:

$$\Delta T(t) = \frac{E(t)}{mc} = \frac{V^2}{4mcR} \left(t + \frac{RC}{4} \left(4e^{-2t/RC} - e^{-4t/RC} - 3\right)\right)$$

6. Equivalent resistance calculations:

$$R_{eq1} = \frac{8 \times 4.8}{8 + 4.8} = 3 \text{ k}\Omega \quad (25)$$

$$R_{eq2} = \frac{1}{\frac{1}{12} + \frac{1}{6} + \frac{1}{4}} = 2 \text{ k}\Omega \quad (26)$$

$$R_{eq3} = 1000 + R_{eq1} + R_{eq2} = 6 \text{ k}\Omega \quad (27)$$

$$R_{eq} = \frac{2 \times 6}{2 + 6} = 1.5 \text{ k}\Omega \quad (28)$$

$$V_{in} = R_{eq} I_{in} = 1.5 \times 20 = 30 \text{ V} \quad (29)$$